

ARCL Instruments Private Limited

Complete Platform Documentation & User/Admin Operations Manual • Version 2.0 (2026)

1. System Overview & Technology Architecture

ARCL Instruments Private Limited is an enterprise-grade digital platform designed for the manufacturing, supply, testing compliance, catalog distribution, and NABL calibration of civil engineering, concrete, soil, bitumen, aggregate, cement, and material testing laboratory equipment.

Core Technology Stack Architecture

```
' Frontend: Next.js 16 (App Router) + React 19 + Turbopack + Tailwind CSS v4
' Backend: Node.js + Express.js REST API with JWT Auth, Role-Based Access Control (RBAC)
' Database: MongoDB Atlas (M0 Cluster) with high-performance compound indexing
' Media Storage: Cloudinary Global CDN with Base64 QR code fallbacks
' SEO & Performance: JSON-LD Schemas, Turbopack Pre-rendering, Dynamic Sitemap, Caching
' Hosting: Vercel Edge Global Network (Frontend) + Render Cloud Server (Backend)
```

2. User Storefront Walkthrough (Customer/Visitor View)

The public website is crafted for civil engineers, laboratory technicians, quality managers, contractors, and universities across India and globally.

Page-by-Page User Experience:

• Homepage (/):

Features hero animated slides, primary testing categories, top featured machines showcase, compliance matrix (IS/ASTM/BS), dedicated regional service hubs (Mumbai, Navi Mumbai, Thane), and interactive FAQs.

• Product Catalogue (/products):

Allows visitors to filter equipment by Category and Equipment Type, search with fuzzy keyword matching, view live status, and instantly trigger direct quotation requests.

• Category Hubs (/categories/[slug] & Short URLs):

Displays category-specific equipment with dynamic filters (Capacity, Automation grade, Motor type), general technical specifications, and working principles.

• Product Detail Page (/products/[slug]):

Presents high-res image zoom gallery, SKU/Product Code, HSN code, technical parameters table, complete set includes accessories, applications, 1-Click WhatsApp quote generator, and PDF catalog downloader.

• Testing Standards Directory (/standards):

Interactive matrix linking Indian Standards (IS 516, IS 2720, IS 2386, IS 1208, IS 4031) and ASTM codes directly with certified machine packages and 1-Click BOQ downloads.

• Calibration Services (/calibration-services):

Enables laboratories to request multi-point NABL traceable calibration, load cell calibration, and on-site machine commissioning certificates.

• Regional SEO Hubs (/civil-lab-equipment-mumbai /navi-mumbai /thane):

Dedicated commercial landing pages optimized for regional contractors, infrastructure projects, and local delivery in Mumbai Metropolitan Region.

• Technical Knowledge Blog (/blog & /blog/[slug]):

Provides laboratory test procedure walkthroughs, calculation formulas, QA/QC checklists, and standard code summaries.

• AI Technical Lab Assistant (Chatbot):

Floating interactive AI assistant available on all storefront pages, answering technical equipment specifications, IS code requirements, and instantly capturing quotation leads.

3. Admin Panel & Operations Manual (Admin View)

The Admin Panel is accessible at /admin/login with enterprise security, role-based access control, active presence tracking, and full management capabilities.

Admin Modules & Controls:

• 1. Dashboard (/admin):

Real-time overview of total products, inquiries, categories, active users, error logs, and visitor trends with instant notification indicators.

• 2. Equipment Types (/admin/equipment-types):

Manage high-level groupings (e.g. Concrete, Soil, Bitumen). Features Drag-and-Drop visual reordering, Featured on Home toggle, and Added Date & Time tracking.

• 3. Categories Management (/admin/categories):

Create and configure product categories, define custom dynamic filters, set general specifications, describe working principles, and view quick details modal with exact timestamp.

• 4. Products Inventory (/admin/products):

Full CRUD operations for testing instruments, multi-image upload via Cloudinary, SKU & HSN code management, dynamic specification inheritance, bulk QR code generation, and Added Date & Time tracking.

• 5. Inquiries & Quotation Leads (/admin/inquiry):

View, filter, and manage incoming buyer inquiries (Name, Phone, Email, Company, State, Message, Product). Track status (New, Contacted, Quoted, Closed) and export to CSV.

• 6. Contact Messages & Subscribers (/admin/contact-messages & /subscribers):

Manage incoming general contact submissions and newsletter subscriber emails with instant export capability.

• 7. Technical Blogs CMS (/admin/blogs):

Publish technical test guides, edit articles with rich formatting, configure custom SEO metadata, upload banner images, and control draft/published status.

• 8. System Error Logs (/admin/error-logs):

Real-time monitoring of client-side and server-side runtime errors, stack traces, and system health.

• 9. User & Staff Management (/admin/users):

Manage team members with Role-Based Access Control (Super Admin, Admin, Staff) and granular permissions per module.

4. Step-by-Step Operator How-To Guide

How to Add a New Product to the Storefront

- ' Step 1: Navigate to Admin Panel -> Products -> Click "+ Create New Product"
- ' Step 2: Enter Product Name, SKU / Product Code (e.g. ARCL-CTM-2000), and HSN Code
- ' Step 3: Select Equipment Type and Category (specs and filters automatically load)
- ' Step 4: Upload high-resolution product photos (drag & drop up to 5 images)
- ' Step 5: Fill technical specifications, features, applications, and "Complete Set Include"
- ' Step 6: Toggle "Featured Item" if you want it showcased on the Homepage
- ' Step 7: Click "Save Product" -> Product goes live instantly on the website!

How to Generate QR Codes for Physical Machines

- ' Step 1: In Admin Panel -> Products Inventory -> Click "Generate All Missing QR Codes"
- ' Step 2: The system generates high-resolution direct-link QR codes for every product
- ' Step 3: Click the QR icon on any product to view, print, or attach the QR label to physical machines

How to Reorder Homepage Showcase Sections

- ' Step 1: Navigate to Admin Panel -> Equipment Types
- ' Step 2: Click and hold the grip icon ((?)) on any row and drag it up or down
- ' Step 3: Alternatively, click the Up (%²) or Down (%¹/₄) arrow buttons
- ' Step 4: The order saves automatically and immediately updates on the live Homepage!